

EVAL Cane: An IoT based Smart Cane for the Evaluation of Walking Gait and Environment

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Abstract—The usage of Internet of Things (IoTs) in healthcare has increased sharply recently. The IoT technology is identified as a potential solution to alleviate the pressures on healthcare systems. This paper proposed a low-cost, non-intrusive and multi-function IoT based smart cane that can be used for the evaluation of human walking and environment, *i.e.*, EVAL Cane, for the elderly people or fragile people. Several sensors such as accelerometer/gyroscope, ultrasonic sensor, force sensitive resistor (FSR), and temperature sensor were instrumented on the cane to collect the data of user's walking gait and walking environment, which will be connected to the Internet through communication protocols like LoRa or WI-FI. Although these sensors were not installed on the human body directly, the user's gait feature can be described by certain of them such as accelerometer and FSR, that means the user's gait can be analyzed and evaluated during the walking with the cane. Besides, based on the data analysis of accelerometer, we can also detect unwanted events such as falls during the walking, which is helpful for the monitoring of user's health status. In addition, the cane has an ultrasonic sensor used to detect an obstacle in front of the cane and this sensor will work together with a buzzer in order to give a sound signal to the user when an obstruction is detected in front of him/her. The buzzer will also make a sound when a fall is detected. Moreover, the environmental temperature of the cane can be measured and be displayed on a LCD screen to the user along with the number of steps taken and the battery percentage level of the device. In a word, EVAL Cane is capable to monitor user's walking, detect the surrounding environment, give output back to the user, and collect data used for further analysis.

Index Terms—Internet of Things (IoTs), smart cane, gait analysis, health monitoring, elderly

I. INTRODUCTION

Nowadays, we enter in the era of Internet of Things (IoTs), which is beginning to shape the future of many industries. Specifically, the usage of IoTs in personal healthcare has increased sharply recently, and 87% of healthcare organization have adopted IoT technologies by 2019 [1]. This is because IoTs have the power to transform the healthcare industry and the IoT technology is identified as a potential solution to alleviate the pressures on healthcare systems [2]. To this end, many IoT based connected devices were developed for remote health monitoring and tracking of elderly. For example, WeWALK is a connected cane that can be paired with smart

phone to get navigation and also be integrated with Voice Assistant and Google Maps [6].

The use of a walking cane is common among elderly people or fragile people who have lower back or hip pain and also people who have trouble seeing. Normally in these type of situations users will use the cane for a long period of time in some cases, even for the rest of their lives. In order to develop more functions on the cane for users, their doctors or guardians, many smart canes instrumented with sensors were developed to satisfy users' different needs. For example, [3] proposed a sound and touch based smart cane. The design was incorporated with Bluetooth enabled obstacle detection module, supported with heat detection and haptic modules. The ultrasonic range finders helped in detecting obstacles. Similarly, [4] also developed a smart cane with distance measurement system for blind people. An ultrasonic sensor was used to measure distance from the obstacle and an earphone was used to alert it. In order to alert obstacles more effectively, [5] added a model-based state-feedback control in smart cane.

The above studies about smart cane focus on detection of obstacles, alert output, navigation and so on. However, for the cane users such as the elderly people or the fragile people who had difficulty walking, these functions are not sufficient. What they need may be gait analysis, which can offer walking quality indicators for their doctors to evaluate their level of mobility, even health state because some maladies can influence the gait feature such as Parkinson's disease. In our previous studies [7], [8], a rollator walker was instrumented with encoders and a 3D accelerometer/gyroscope to analyze the walking of young and elderly people. The results showed that users' steps could be detected by using the zero-pass moments of angular velocity collected from the accelerometer/gyroscope. Thus, many usual walking gait indicators such as step number, stride length and walking speed were obtained. This means it is possible to do gait analysis by using this kind of device instrumented with sensors. In other word, it is possible to evaluate users' walking level and monitor their health state. Moreover, since the sensors were not installed on the human body directly but on a mobility device which could be used everyday by the elderly, this method is not intrusive and low-cost for users.

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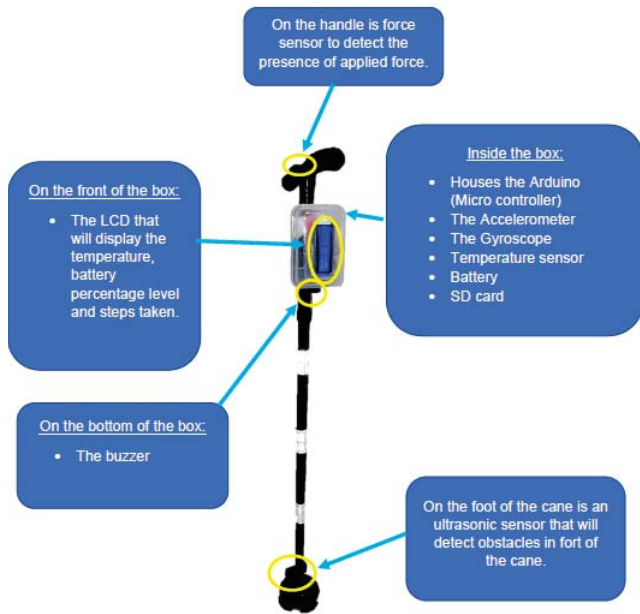


Fig. 1. The EVAL Cane.

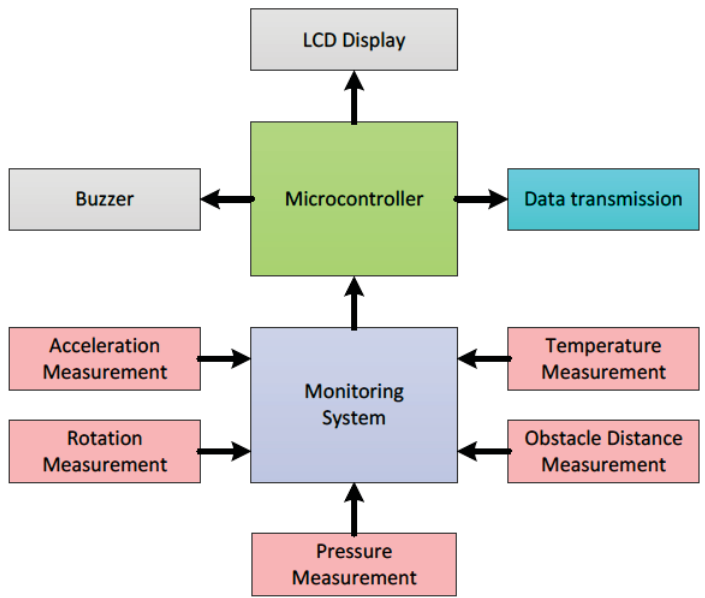


Fig. 2. The functions of EVAL Cane.

Following the same objective, in this paper we designed an IoT based connected EVAL Cane, which is also installed a 3D accelerometer/gyroscope for gait analysis. Besides, other functions such as detection of falls, detection of obstacles, alert output, evaluation of walking environment, and remote monitoring are also realized on our connected EVAL Cane.

II. DESCRIPTION OF CONNECTED EVAL CANE

This smart cane is designed based on an original frame but with a few extra features added to it, such that we make the cane capable of monitoring its utilization and environment. As shown in Fig. 1, the inputs of the whole system, *i.e.*, the sensors installed on the cane, are an accelerometer/gyroscope, a temperature sensor, an ultrasonic sensor and a force sensitive Resistor (FSR) respectively. Obviously, our cane equipped with FSR can collect more information of user's walking compared with other smart canes that only equipped with accelerometer or ultrasonic sensor. The FSR is a resistor that changes its resistance if pressure is put on it. When there is no force applied on the FSR, the resistance is extremely high acting as an open circuit. When a pressure is put on it, the resistance will decrease. The larger the pressure is, the less is the resistance. Here FSR is installed on the top of the cane's handle so it can detect if users acted on the cane and how much pressure was put, which is helpful for the gait analysis.

For the outputs of the cane, we installed a small LCD Display and a Buzzer at present. The LCD Display was equipped on the top of the cane in order to offer some information of the cane, user's walking, and the walking environment. Now it can show the battery percentage level, the number of steps finished and the temperature. However, which information will be displayed on LCD can always be reset according to the needs of users. In addition, another output is

a Buzzer. It works when an obstacle or a falling is detected, and two different voices are set for the two cases respectively.

To collect the sensors' data and do further analysis, all the sensors were connected with Arduino UNO as a microcontroller. All the measurement that it takes will automatically be updated in its internal data registers. In summary, a block diagram used to describe the functions of our cane is shown in Fig. 2.

For the sake of remote walking monitoring and tracking, our EVAL cane can be directly connected to Internet leveraging communication protocols like LoRa or Wi-Fi. This can be achieved by the LoRa and Wi-Fi interfaces, which are directly integrated in the Arduino card. Hence, the walking data collected by the connected EVAL cane can be then transmitted to server in real time for online visualization, health monitoring and further analysis (disease prediction, authentication, walking ground evaluation or recognition).

III. ILLUSTRATIVE EXPERIMENTAL RESULTS

We tested the cane and confirmed that all the expected functions such as detection of falls and obstacles, alert output, and gait analysis, can be realized. The EVAL cane was installed with five sensors, and each sensor's data was recorded with a fixed time interval (230 ms). The temperature sensor is used to obtain the temperature which will be shown in an LCD. The ultrasonic sensor data is used to detect obstacles. The safe distance is configurable in the program of the microcontroller, *e.g.*, 50 cm. If the distance from the cane to the obstacle becomes equal to the safe distance, the Buzzer starts to work. Moreover, if this distance continues to decrease, the closer the distance, the louder the sound. The data of accelerometer, gyroscope, and FSR, which can describe user's gait characteristics directly, are used for fall detection and gait analysis. For

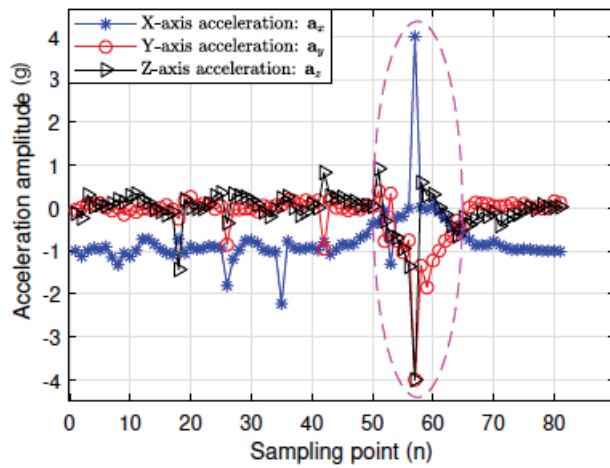


Fig. 3. Data of accelerometer when the cane was suddenly dropped on the ground.

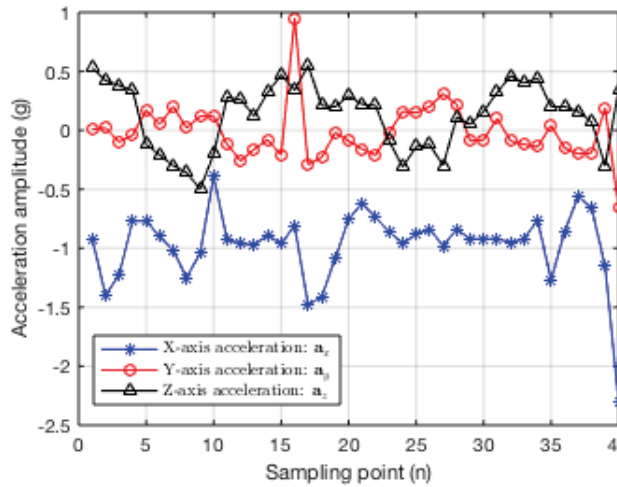


Fig. 4. Data of accelerometer during the walking of a user.

example, we tested the fall detection by using the cane, and the data of accelerometer when the cane was suddenly dropped on the ground is shown in Fig. 3. It can be seen that there is an immediate and large change on the data of accelerometer. In the microcontroller, this change is regarded as a fall and the Buzzer will be activated. Next, when the cane is re-centered, the acceleration data returns to normal. The accelerometer and gyroscope are both 3-axis sensors, so their data includes X-axis, Y-axis, and Z-axis directions data respectively. In addition, since the accelerometer and gyroscope are mounted on the pole so that the X-axis represents the direction opposite to gravity, as shown in Fig. 3. This is why the value of the acceleration on the X-axis (*i.e.*, a_x) equals approximately -1 gravity during normal gait.

In order to do gait analysis, an experiment of walking with EVAL cane was designed. The subjects were asked to walk at their normal speed and along a straight line of 10 m with

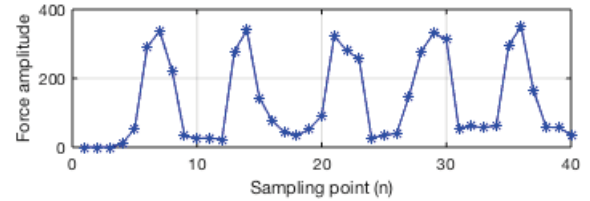
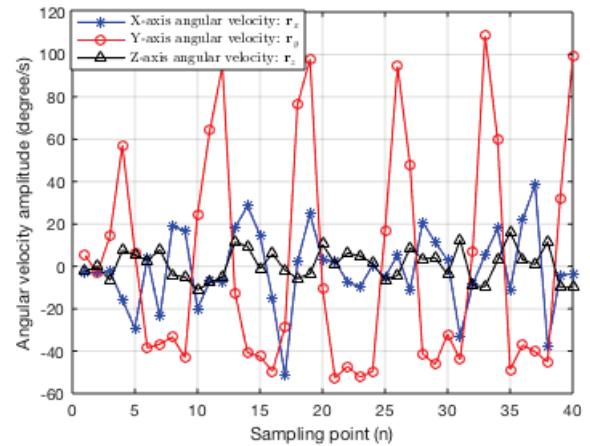


Fig. 5. Data of gyroscope and FSR during the walking of a user.

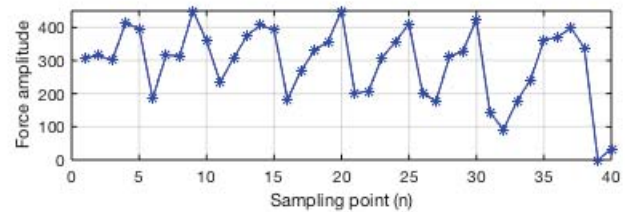
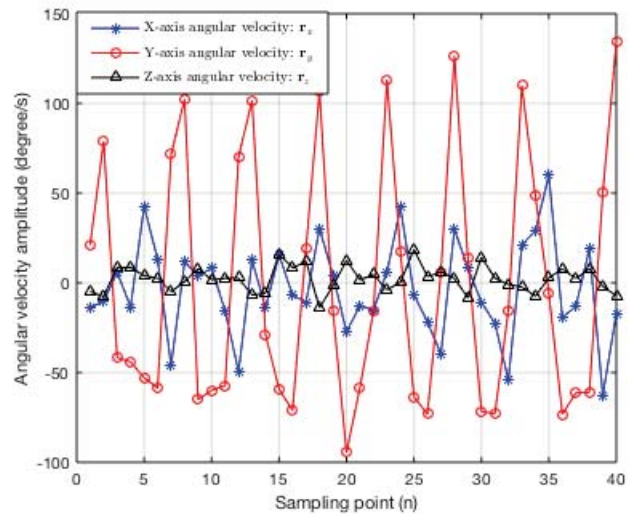


Fig. 6. Data of gyroscope and FSR during the walking of another user.

a flat road surface. In addition, they were given 10 minutes to get familiar with walking with the EVAL cane before the formal experiment. Fig. 4 and Fig. 5 show one user's data of accelerometer/gyroscope and FSR during the walking of one test. We can clearly see that the pattern of FSR data and angular velocity are cyclic, which can be used to detect the user's steps precisely. In fact, since the user only applies force on the handle of cane for one support leg during the walking, one cycle of the FSR data means that the amount of steps finished by the user is two. In addition, the walking quality indicators can be calculated or estimated by using the data of accelerometer, and the same method has been presented in our previous studies [7], [8]. Moreover, the results of another subject in the same condition were shown in Fig. 6. It shows that there exists significant difference in the FSR data of two users. Several factors can lead to this result, *e.g.*, users' height, walking habits, health status, physical fitness, etc. Therefore, in the next step it is interesting to verify if the FSR data is a mobility-related indicator which could be used to evaluate user's mobility or it is just influenced by user's physical characteristics. In order to do that, many subjects with different age, sex, height, disability and reduced mobility should be considered in the experiments.

IV. CONCLUSIONS

This paper proposed a multi-function smart cane for the fragile people with reduced mobility. Based on the installed sensors, it can realize the assistance functions such as obstacle detection, fall detection and emergency warning. Moreover, it can collect users' walking data such as acceleration, angular velocity, and pressure acting on the handle of the cane, which can be used to analyze users' gait and monitor their health status indirectly. For the fragile people who need to use a cane in his/her daily life, our cane is non-intrusive compared with some wearable monitoring devices such as smart watch or shoes. Furthermore, since the instrumented cane can collect both the data of human and environment, these information could be analyzed and used to develop other functions such as gait identification, ground evaluation and so on.

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